They Like to Move It, Move It — Determining the Viscosity of Colloidal Suspensions of Monodisperse Polystyrene Microspheres in Water Using Brownian Motion

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Brownian motion is the phenomenon of microscopic particles suspended in fluid moving randomly. We use the observation of particles undergoing Brownian motion to determine the viscosity of a colloidal suspension. We use a suspension of monodisperse polystyrene microspheres in water to track movement of the particles and through that determine the coefficient of diffusion of our sample. For the particloe tracking we use the python package trackpy. Using the Stokes-Einstein relation given by $D = \frac{k_B T}{6\pi\eta r}$, where D is the diffusion coeffcient in m^2/s, $k_B = 1.38 \times 10^{-23} J/K$ is the Boltzmann constant, T is the temperatur ein Kelvin, η is the dynamic viscosity in $mPa \cdot s$ and r is the radius of the particle, we then relate the diffusion coefficient to the viscosity. We measured the Diffusion coefficient of 25 videos across 4 different particle radii $r \in \{0.373, 0.53, 0.77, 1.03\}\mu m$. We found a mean value of viscosity for the different aprticle radii as $\eta_{r=0.373} = 2.7 \pm 0.1 mPa{\cdot}s, \eta_{r=0.53} = 2.1 \pm 0.2 mPa{\cdot}s, \eta_{r=0.77} = 2.2 \pm 0.2 mPa{\cdot}s, \eta_{r=1.03} = 2.0 \pm 0.4 mPa{\cdot}s$. Perfroming a constant function fit through the data yielded a viscosity of $\eta = 2.4 \pm 0.1 mPa \cdot s$. The difficulties of the experiment and subsequently errors in measurements were influenced by inability to dierctly measure sample temperature as only room temperature measurements were performed, incosnsistent sample lighting, sample interference that created phase contrast transitions and forced us to de-focus the samples to prevent change from dark to light colour, and finally the tracking of the particles.

I. INTRODUCTION

Brownian motion describes the continuous, random movement of microscopic particles suspended in a fluid. The phenomenon was first observed by Robert Brown in the nineteenth century, who noticed the irregular movement of pollen grains suspended in water. Its theoretical foundations were subsequently developed by Albert Einstein providing a physical explanation for the observed motion and establishing its connection to the microscopic properties of fluids [1].

Einstein's theoretical description relates the mean squared displacement of suspended particles to the diffusion coefficient and elapsed time [2]. This relationship provides a way to investigate the properties of a fluid through the motion of particles suspended within it. In particular, the Stokes-Einstein relation relates the diffusion coefficient to the particle size, fluid viscosity and temperature [3]. Consequently, by tracking the Brownian motion of particles with known properties, the viscosity of the surrounding fluid can be determined.

In this experiment, Brownian motion is used to determine the viscosity of aqueous suspensions containing monodisperse polystyrene microspheres of different sizes. By tracking the displacement of individual particles over time, their diffusion coefficients can be determined and related to the viscosity of the surrounding fluid. The main objective is to investigate whether this method provides a reliable determination of viscosity. If successfull, this would allow an *in situ*[4] measurement of the viscosity of the liquid. This is especially advantageous since it allows for measurements without disturbing the microsystem [5].

Since we are measuring the viscosity of an aquatios solution, we can compare our results to the known viscosity of water at room temperature, which is approximately $1.002\,\mathrm{mPa \cdot s}$ at $20°C$ [6]. This comparison will help us assess the accuracy and reliability of our measurements and provide insight into the effectiveness of using Brownian motion as a method for determining fluid viscosity.

II. THEORY

A key relation connecting Brownian motion to the fluid's viscosity is the Stokes–Einstein relation, which is given as follows

$$D = \frac{k_B T}{6\pi\eta r}, \tag{1}$$

where D in $\mathrm{m^2/s}$ is the diffusion coefficient, k_B is the Boltzmann constant given as $1.38 \times 10^{-23}\,\mathrm{J/K}$, T is the thermodynamic temperature in Kelvin, η is the dynamic viscosity in Pascal-seconds, and r is the particle radius in meters [2]. It applies to spherical particles suspended in a Newtonian fluid and provides us with a way to determine the viscosity of the fluid by measuring the diffusion coefficient of the suspended particles. These conditions are mostly fufilled by our setup. We can rewrite this equation in terms of the viscosity as

$$\eta = \frac{k_B T}{6\pi D r}. \tag{2}$$

To get the diffusion coefficient as a function that we can measure, we look into the movement of particles that

undergo Brownian motion. To begin, we assume that the motion of each distinct particle is independent of the motion of the other suspended particles. If we then also impose the condition that the number of particles per unit volume ν is a function of only x and t, i.e. $\nu = f(x,t)$, we get that it sattisfies the equation

$$\frac{\partial f}{\partial t} = D\frac{\partial^2 f}{\partial x^2}, \tag{3}$$

where D is the diffusion coefficient [2].
We then define $f(x,t)dx$ to denote the number of particles that have moved by a displacement between x and $x+dx$ in the x direction in the time interval $[0,t]$. We can impose the initial and boundary conditions as

$$f(x,0) = 0,$$
$$\int_{-\infty}^{\infty} f(x,t)dx = n,$$

where n is the total number of particles suspended in the liquid [2]. The solution to $f(x,t)$ can in this case be found analytically and is given by

$$f(x,t) = \frac{n}{\sqrt{4\pi Dt}} e^{-\frac{x^2}{4Dt}}. \tag{4}$$

We can then determine the average displacement of a particle along the x direction as

$$\lambda_x = \sqrt{\langle x^2 \rangle} = \sqrt{2Dt}, \tag{5}$$

where $\langle x^2 \rangle$ is the arithmetic mean of squares of displacements along the x direction [2].
In the 2D case, we take the effect to be independent in the x and y axes. This then gives us an expression for the mean squared displacement in the xy-plane as

$$\langle R^2 \rangle = \langle x^2 \rangle + \langle y^2 \rangle = \langle x^2 + y^2 \rangle = 2Dt + 2Dt = 4Dt. \tag{6}$$

We can then use this relation to find an equation for the diffusion coefficient as

$$4Dt = \langle R^2 \rangle, \tag{7}$$
$$\frac{d}{dt}4Dt = \frac{d}{dt}\langle R^2 \rangle, \tag{8}$$
$$4D = \frac{d}{dt}\langle R^2 \rangle, \tag{9}$$
$$D = \frac{d\langle R^2 \rangle}{4dt}. \tag{10}$$

We can then plug this expression into Eqn. 2 to get

$$\eta = \frac{k_B T}{6\pi D a} = \frac{2k_B T}{3\pi \frac{d\langle R^2 \rangle}{dt} a}. \tag{11}$$

Calculating the uncertainty of a quantity is done using the formula

$$\sigma_A = A \cdot \sqrt{\sum_n \sigma_n^2}, \tag{12}$$

where A represents the quantity we want to calculate the resulting ucnertainty for and σ_n are relative uncertainties of quantities that are in the formula for A.

III. METHODOLOGY

A schematic of the experimental setup is shown in Figure 1.

As samples to be analysed, we used monodisperse polystyrene microspheres with factory-specified diameters of $0.75 - 2\mu$m, suspended in water [7]. The resulting liquid is a colloidal suspension, which describes mixtures in which the colloidal particles are significantly larger than the solvent molecules but small enough to exhibit Brownian motion [8]. In our case, the particles are suspended at a concentration significantly lower than that of the stock suspension in which they are supplied, namely a 2.5% solids (w/v) aqueous suspension [7]. We used samples with a mass-per-volume concentration of 0.1% across all measurements to minimise the effects of concentration as an additional variable and to keep the results comparable. These samples were prepared by diluting the stock suspension with water.

We prepared samples with four different particle radii, namely $r \in \{0.373, 0.53, 0.77, 1.03\}\mu m$, for which the differences from the factory-specified values had been determined in advance. The samples were then placed on a microscope slide and covered with a coverslip to prevent evaporation of the liquid. The slide was placed under a microscope with a magnification of $40\times$, and a light source was positioned at an angle to the sample to create contrast and make the particles more visible.

To calculate the final viscosity using Eqn. 11, we needed to determine the temperature and diffusion coefficient. The temperature was measured using a room thermometer located in the room where the experiment was conducted. This thermometer had a precision of $0.1°C$.

We calculated the diffusion coefficient by measuring the mean squared displacement $\langle R^2 \rangle$ of the particles over time. To do this, we recorded videos of the particles using the DinoCapture 2.0 software. The videos were recorded at a frame rate of 20 frames per second, with durations between 3 and 9 seconds and a resolution of 640x480 pixels. The resulting videos were then processed using the Python library Trackpy to track the movements of the particles and calculate their mean squared displacement over time.

For the tracking to work correctly, several parameters had to be tailored to each video. One required parameter was an estimate of the particle size in pixels, called the "feature size". This value was mostly kept uniform across videos of particles with the same diameter. In some cases, however, the microscope focus required the value to be adjusted for Trackpy to track the particles correctly. We identified this as an apparent diffraction effect, in which a bright spot appeared in the middle of the particle due to interference. This meant that a higher feature size was needed to track the entire particle and avoid tracking only part of it. Another important parameter was the so-called "minmass". This parameter determines how strong the particle signal must be to be detected and tracked. By setting a higher minmass, par-

ticles had to be more clearly visible and in the foreground to be tracked. This helped eliminate accidental tracking of shadows from out-of-focus particles, as well as dust present in the setup. To identify these particles, we visually inspected the tracking results and adjusted the parameters until the tracking was clear throughout the video. We also implemented drift correction to eliminate systematic particle movement caused by external factors, such as vibrations or air currents. This was done by calculating the average displacement of all tracked particles and subtracting it from the individual particle displacements.

To convert the tracked mean squared displacement from pixels to micrometres, we needed to determine a calibration factor expressing how many micrometres corresponded to one pixel. For this, we used the software Fiji and a micrometer ruler placed under the microscope with the same magnification and camera settings as those used for our videos. We then took a picture and measured the distance between eight marks on the ruler in pixels, while the corresponding physical distance was known. We estimated the uncertainty from the width of one mark, which yielded a calibration factor of $0.130 \pm 0.003 \mu m/px$.

The mean squared displacement (MSD) was calculated from the tracked particle trajectories using the `emsd` function from the Python library Trackpy. This function calculates the ensemble mean squared displacement by averaging squared particle displacements over the available trajectories and time origins for each time lag. The pixel-to-micrometre calibration factor and the video frame rate were supplied to convert the resulting displacements into physical units and time. The maximum time lag was set to one quarter of the total number of video frames to limit the analysis to time intervals with sufficient tracking data. The resulting MSD values were then used to determine the diffusion coefficient from a linear fit.

To obtain the diffusion coefficient from the mean squared displacement, we used the formula in Eqn. 10. We fitted a linear function to the mean squared displacement as a function of time to obtain a slope (derivative) equal to $4D$. We constrained the fit to pass through the origin, in accordance with the theoretical expectation that the MSD is zero at zero time lag. The diffusion coefficient was then obtained by dividing the fitted slope by four. This constraint assumes that any experimental offset in the MSD is negligible; its possible influence on the fitted diffusion coefficient should therefore be considered when interpreting the results.

The uncertainties propagated through this process included the uncertainty in the particle diameter, the uncertainty arising from the thermometer precision, the uncertainty in the micrometre-to-pixel conversion, and the uncertainty in the diffusion coefficient obtained from the linear fit. The uncertainty in the viscosity, as well as all preceding uncertainty propagations, was calculated using the formula in Eqn. 12.

Once the diffusion coefficient had been calculated, we used the Stokes-Einstein relation (Eqn. 11) to determine the resulting viscosity for a given particle radius and its uncertainty. To average the different viscosity values calculated for the same radius, we used the Birge ratio method to account for a possible mismatch between the individual uncertainties and the spread of the viscosity values [9].

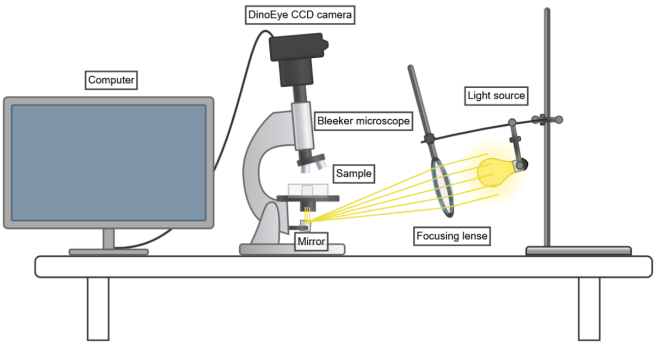


FIG. 1. Schematic of the experimental setup used to record the Brownian motion of polystyrene microspheres suspended in water. The sample was placed on a microscope slide under a 40× microscope objective and illuminated from an angle to improve particle contrast. Particle motion was recorded using a camera connected to a computer.

IV. RESULTS

The measured room temperature ranged from $20.9°C$ to $21.0°C$. The diffusion coefficients obtained from the particle trajectories were of the order of $10^{-1}\,\mu\text{m}^2/\text{s}$, depending on the particle radius. In total, we obtained 25 viscosity estimates. These were grouped according to particle radius: 6 measurements for $r = 0.373\,\mu\text{m}$, 6 for $r = 0.53\,\mu\text{m}$, 8 for $r = 0.77\,\mu\text{m}$, and 5 for $r = 1.03\,\mu\text{m}$. We have summarised the averaged values for the viscosity obtained for each particle radius in Figure 2. The uncertainties were calculated using the Birge ratio method to account for any discrepancies between the individual uncertainties and the observed spread of the measurements. The results show that the viscosity values are very different across different particle radii with the final constant function fit not lying within the uncertainty of the individual measurements, which we discuss in Section V. Furthermore, the reduced chi-squared value of the constant function fit was $\chi^2_\nu = 7.87$, clearly indicating that the spread of the data points is larger than expected based on the individual uncertainties. A comparision with the textbook value of the viscosity of water at room temperature, which is approximately $1.002\,\text{mPa}\cdot\text{s}$ [6], shows that our measured values are significantly higher than the expected value.

An example of the fitting of the mean squared displacement to obtain the diffusion coefficient is shown in Figure 3. It becomes clear that the linear fit and the uncertainties are valid, with a reduced chi-squared value of

$\chi^2_\nu = 1.005$. This data is exemplary for the other measurements which had similar reduced chi-squared values.

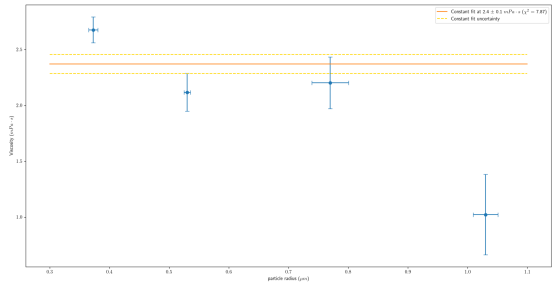


FIG. 2. Average viscosity obtained for each particle radius. The points represent the average viscosity values for different particle radii. The solid line shows the constant fit, with the dashed lines indicating the uncertainty of the fitted value.

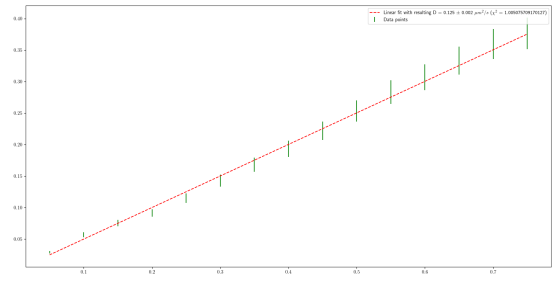


FIG. 3. Example of the mean squared displacement (MSD) as a function of time for the sample with particle radius $r = 0.53\,\mu$m of the measurement A003 - 20261021_141507. The green data points represent the calculated MSD values, while the dashed red line shows the linear fit constrained to pass through the origin. The slope of the fit was used to determine the diffusion coefficient.

V. DISCUSSION

Although our measurements demonstrate that the viscosity of a liquid can be determined using Brownian motion, the obtained values do not agree with the expected viscosity of water. This suggests that systematic errors affected the accuracy of our measurements. Similar experiments performed by other groups have yielded values closer to the expected viscosity of water [5]. We believe that the main source of systematic error in our experiment was the difficulty of reliably tracking the Brownian motion of the particles. Several limitations of our experimental setup may also have contributed to the observed deviations.

The most significant difficulty we encountered was the tracking of particles using Trackpy. Despite adjusting the tracking parameters to minimise incorrect detections, we observed cases in which particles were tracked incorrectly. These errors could not be reliably identified or quantified and were therefore not fully accounted for in our uncertainty estimates. Since the diffusion coefficient is determined from the mean squared displacement of the tracked particles, incorrect particle positions can directly affect the calculated diffusion coefficient and, consequently, the resulting viscosity. We therefore consider tracking errors to be a likely major contributor to the systematic deviation in our results.

One factor that made particle tracking particularly difficult was the optical behaviour of the particles around the focal plane. We observed that particles could transition from dark to bright as they moved through the focus, which made their appearance inconsistent between frames. This complicated the identification of the same particle across consecutive frames. To reduce this effect, we deliberately introduced a slight defocus when recording the particles. Although this made the particles less sharply defined, we found that their appearance became more consistent and that the Trackpy parameters could be adjusted more easily. We therefore considered slightly defocused particles more suitable for tracking, although we did not quantitatively compare the tracking accuracy under focused and defocused conditions.

In addition to the tracking difficulties, the experimental setup was not fully reproducible between measurements. The stands holding the light source and lens were not fixed relative to the microscope. Consequently, disassembling and rebuilding the setup changed the lighting conditions. We attempted to reproduce the initial configuration by positioning the components as similarly as possible, but this was done by visual estimation rather than using fixed reference points or positioning tools. A more consistent setup could be achieved by using a microscope with integrated illumination or by mechanically securing the light source to the microscope.

These tracking difficulties were more problematic for smaller particles, which were less sharply defined and more affected by the optical interference effects. This may explain why the viscosity values obtained for smaller particle radii deviated more from the expected value than those obtained for larger particles. It also is plausible that this mainly led to the diffusion coefficient being underestimated, which would lead to an overestimation of the viscosity *ceteris paribus*. This is consistent with the observed trend of higher viscosity values for smaller particle radii.

Another possible source of systematic error was the temperature of the sample. We measured the room temperature using a thermometer, but this did not necessarily correspond to the actual sample temperature. We tried to minimise temperature changes by avoiding unnecessary contact with the sample and the surrounding equipment. However, the light source was directed towards the sample throughout the measurements and may have caused local heating. Since we did not have an in-

strument to measure the sample temperature directly, this effect could not be quantified. Future measurements would benefit from monitoring the sample temperature directly.

Finally, we observed that the calculated viscosity tended to increase when the same sample was observed over a longer period. We suspect that this was related to changes in the sample over time, potentially due to evaporation or other processes affecting the suspension. To reduce this effect, we limited the time for which each sample was used and prepared a new sample when further measurements were required.

Overall, we consider unreliable particle tracking, particularly the changing appearance of particles as they moved through the focal plane, to be the most likely major source of systematic error. Improving the optical conditions and tracking reliability, together with better control of the sample temperature and a more reproducible experimental setup, would therefore be the main priorities for improving the accuracy of future measurements.

VI. CONCLUSIONS

We aimed to determine the viscosity of a colloidal suspension of monodisperse polystyrene microspheres in water using Brownian motion. We tracked the movement of particles in the suspension and calculated their diffusion coefficients, which were then related to the viscosity of the fluid using the Stokes-Einstein relation. Our measurements yielded viscosity values for an aquatic solution of $\eta = 2.4 \pm 0.1\text{mPa} \cdot \text{s}$, which is significantly higher than the expected value of $1.002\,\text{mPa} \cdot \text{s}$ for water at room temperature. The results also showed a trend of higher viscosity values for smaller particle radii, which we attribute to difficulties in tracking the Brownian motion of smaller particles. We identified several potential sources of systematic error, including unreliable particle tracking due to changing particle appearance, inconsistent lighting conditions, and possible local heating of the sample. Future improvements could focus on enhancing tracking reliability, controlling sample temperature more accurately, and ensuring a more reproducible experimental setup. Overall, while our method demonstrated the feasibility of determining viscosity using Brownian motion, further refinements are needed to achieve accurate and reliable measurements. Future research could focus on improving the tracking of Brownian motion, for example by using more advanced image processing techniques or machine learning algorithms to identify and track particles more reliably. If the method is improved, it could provide a valuable tool for measuring viscosity in colloidal systems and investigate the impact of the particle size, concentration, and fluid properties on the accuracy of viscosity measurements.

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